



Lab 07: Pub/Sub Qwik Start - Console

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Overview

Set up a Pub/Sub topic and subscription using the Google Cloud Console. Published a Hello World message to the topic and pulled it using the Cloud Shell command line to confirm delivery.

Key Pub/Sub Concepts

Term	Definition
Topic	A shared channel where publishers send messages
Publisher	Application that creates and sends messages to a topic
Subscription	Named resource representing a stream of messages from a topic
Subscriber	Application that receives messages via pull or push
Pull	Subscriber requests messages from the subscription manually
Push	Pub/Sub automatically sends messages to a webhook endpoint

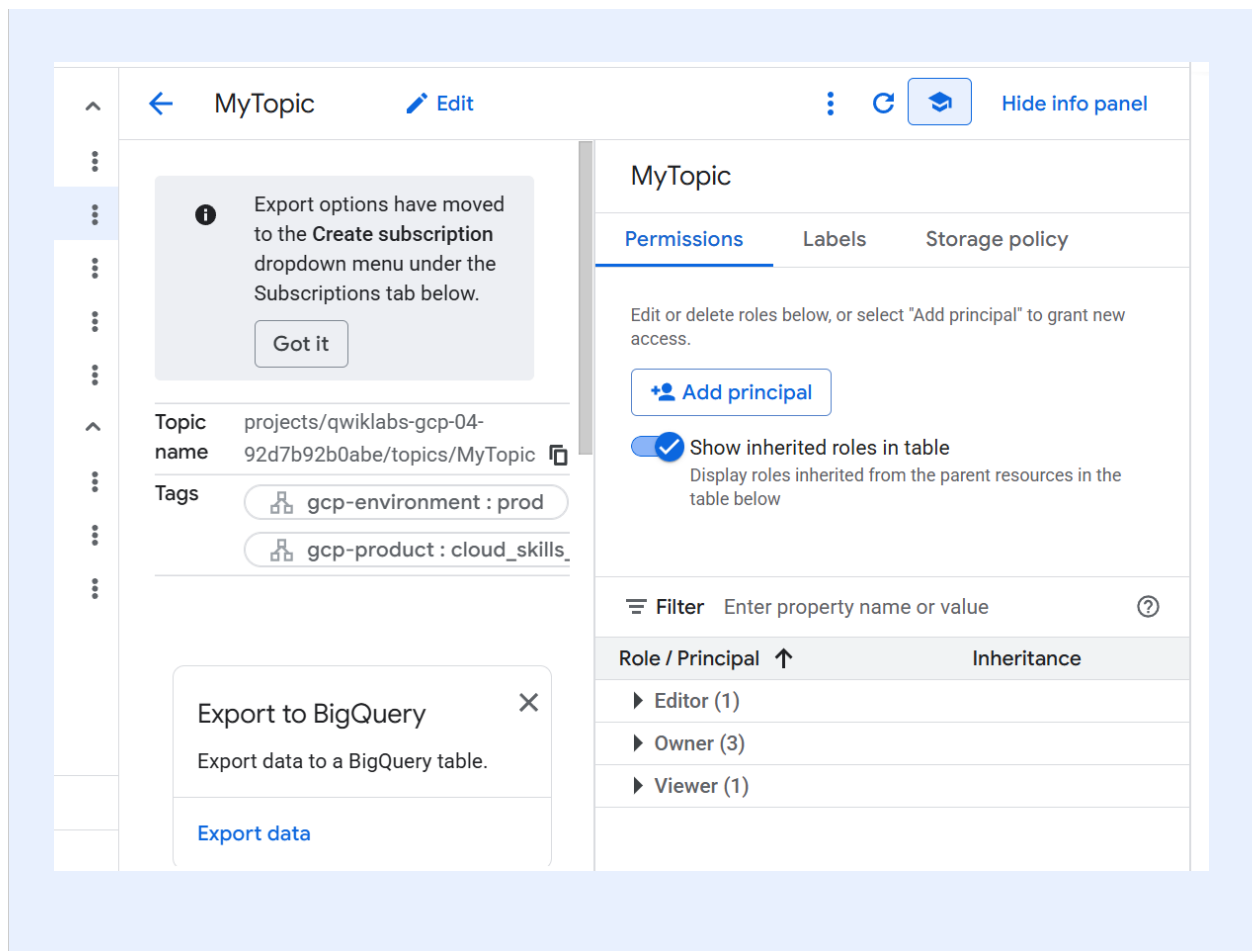
Objectives

- Create a Pub/Sub topic via the console
- Create a pull subscription to the topic
- Publish a Hello World message to the topic
- Pull and view the message using Cloud Shell

Task 1 - Create a Topic

Navigation: Navigation Menu → View All Products → Analytics → Pub/Sub → Topics → Create Topic

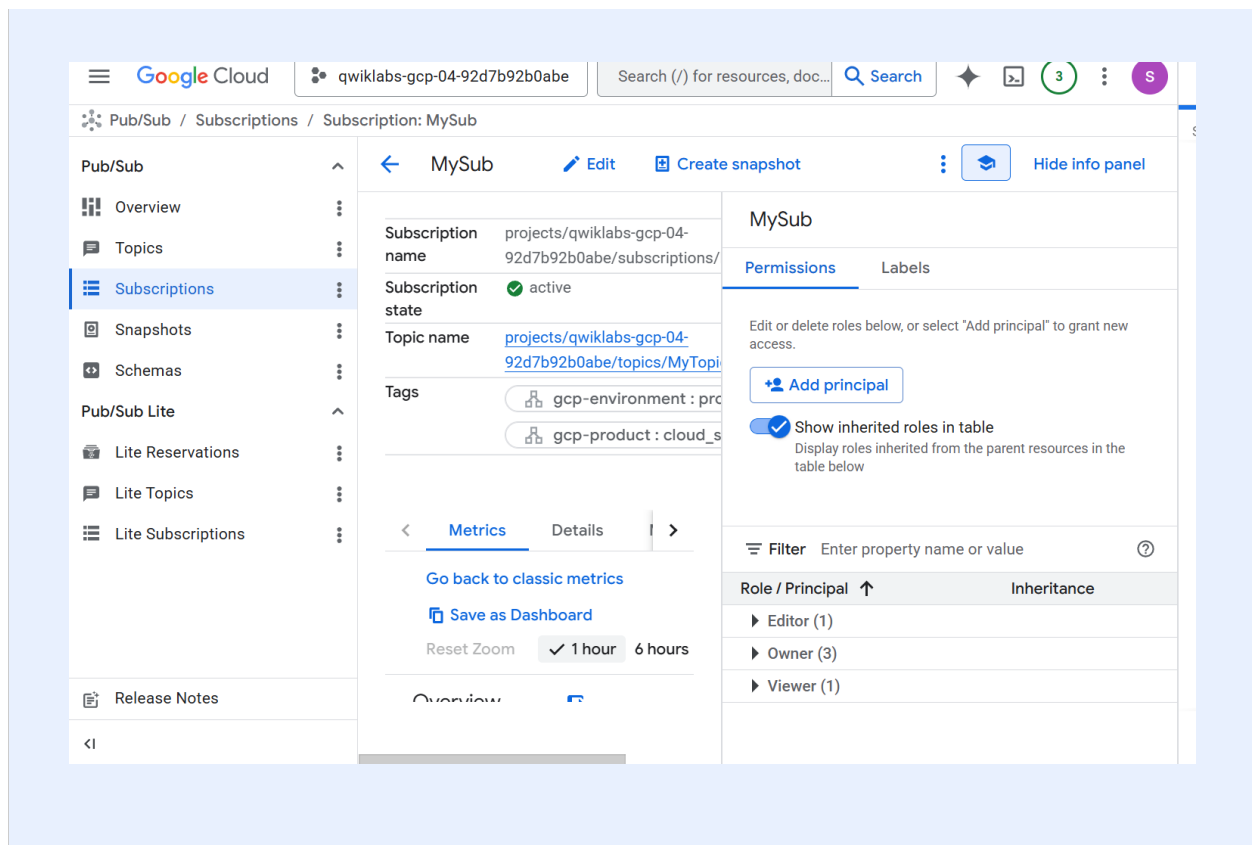
- Topic ID: MyTopic



Task 2 - Add a Subscription

Created a pull subscription named MySub attached to MyTopic.

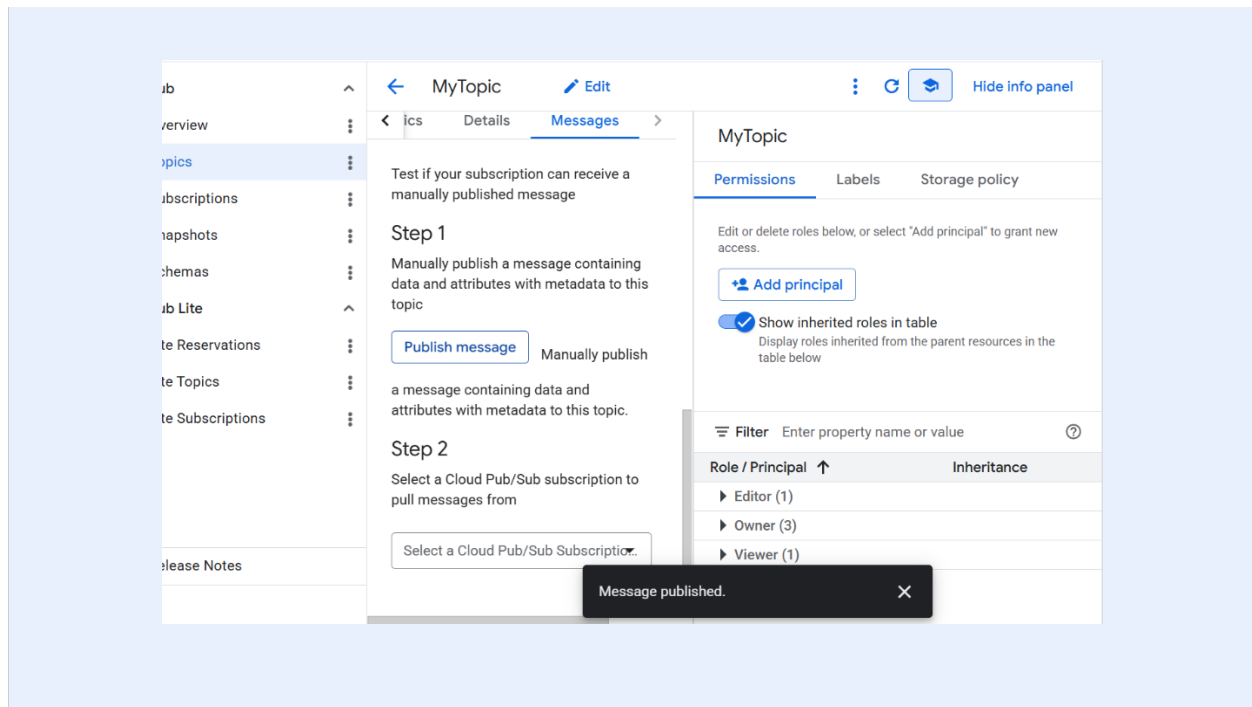
- Subscription name: MySub
- Topic: projects/[PROJECT_ID]/topics/MyTopic
- Delivery type: Pull



Task 3 - Publish a Message

Published a Hello World message to MyTopic via the Messages tab in the console.

- Opened MyTopic → Messages tab
- Clicked Publish Message
- Message: Hello World



Task 4 - Pull and View the Message

```
gcloud pubsub subscriptions pull --auto-ack MySub
```

```
Welcome to Cloud Shell! Type "help" to get started, or type "gemini" to try prompting with Gemini CLI.
Your Cloud Platform project in this session is set to qwiklabs-gcp-04-92d7b92b0abe.
Use 'gcloud config set project [PROJECT_ID]' to change to a different project.
student_01_afdc55dd1197@cloudshell:~ (quiklabs-gcp-04-92d7b92b0abe) $ gcloud pubsub subscriptions pull --auto-ack MySub
DATA: Hello World
MESSAGE_ID: 19624320629040198
ORDERING KEY:
ATTRIBUTES:
DELIVERY ATTEMPT:
ACK STATUS: SUCCESS
student_01_afdc55dd1197@cloudshell:~ (quiklabs-gcp-04-92d7b92b0abe) $
```

Real World Application

- An e-commerce site publishes an order event to a topic when a purchase is made
- Multiple subscribers receive the same event, one sends a confirmation email, one updates inventory, one triggers shipping
- Pub/Sub decouples the systems so they work independently without knowing about each other

What I Learned

- A publisher sends messages to a topic and a subscriber receives via a subscription
- Pull subscriptions require the subscriber to request messages manually
- The --auto-ack flag automatically acknowledges messages after pulling
- Every subscriber must acknowledge each message within a configurable window
- Pub/Sub is asynchronous, senders and receivers are decoupled

Resources

GCP Pub/Sub Docs: <https://cloud.google.com/pubsub/docs>

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